Artificial Intelligence in India's Criminal Justice System: A Case-Based Analysis of Failures, Bias, and the Need for Accountability

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***Abstract*—One of the major problems India is currently facing is AI being used in the Indian judicial system. There are many cases that show the failed usage of AI systems in Indian court cases, arrests, and bail decisions. This paper tells what role AI played in the failure of the system, how it impacted the people involved, and what interventions can be followed to come out of this loop of unfairness and incomplete evaluation of facts. This paper documents, analyses, and categorises ten real cases of AI failure across India's criminal justice and judicial pipeline, identifies the structural reasons these failures keep happening, and proposes concrete solutions that India can actually implement.**

***Index Terms*—Artificial Intelligence, Indian Judiciary, Facial Recognition, Algorithmic Bias, Hallucination, Bail Prediction Fairness, Criminal Justice, AI Governance, CCTNS, Counterfactual Fairness**

I. INTRODUCTION

In March 2020, a man named Ali was arrested in the narrow alleys of Chand Bagh, Delhi. He was arrested not on the basis of any eyewitness testimony, physical evidence, or confession. The decision that led to his arrest was made entirely on the basis of a percentage score produced by a facial recognition algorithm used by the Delhi Police. As a result of his arrest, he spent more than four and a half years in pre-trial detention and was never convicted. When he was finally released, his family was in a debt of twenty-five lakh rupees and his mother had lost almost 40 kilograms of weight.

Ali's case was not the only exception in the failure of the facial recognition technology used by the Delhi Police. Over 750 riot-related cases in Delhi were investigated using the same facial recognition technology. More than 80 percent of those cases have resulted in acquittals or discharges. This means the technology was used at scale, failed at scale, and people suffered at scale [1].

The case discussed above was just one aspect of the whole failure visible across India's jurisdiction and policing system. When looking specifically at the judicial system, the picture becomes equally concerning. In December 2024, a Rs. 669 crore tax tribunal order was recalled because it cited four judgments that do not exist in any legal database [2]. In October 2025, a Rs. 27.91 crore income tax assessment was quashed for the same reason [3]. In August 2025, a trial court judge in Andhra Pradesh dismissed a property case by relying on four fake Supreme Court judgments that were generated by an AI tool [4]. The Supreme Court of India itself, in February 2026, called this trend alarming and issued notices to the Attorney General, the Solicitor General, and the Bar Council of India [5].

Other papers have looked at individual AI tools in isolation. Nobody has looked at the full picture, from policing to arrest to bail to judicial decision, as one connected system of failures. That is what this paper does.

This paper documents, analyses, and categorises ten real cases of AI failure across India's criminal justice and judicial pipeline, identifies the structural reasons these failures keep happening, and proposes concrete solutions that India can actually implement. To understand why these failures are happening, we must first understand what AI systems are currently deployed across India's justice system, what the existing research has found about their biases, and, critically, what the Supreme Court of India itself has already acknowledged about the risks they carry.

II. BACKGROUND AND AI SYSTEMS USED

India is rapidly integrating AI across both its policing infrastructure and its court system. These systems were introduced with the stated goals of reducing case backlogs, improving efficiency, and enhancing the accuracy of law enforcement. As of 2025, India carries over 50 million pending cases, a backlog so large that clearing it at the current pace would take more than 300 years [6]. It is this pressure that has driven judges, lawyers, police officers, and government bodies to reach for AI tools that appear to save time. Understanding what those tools are, and what the existing research says about them, is the necessary foundation for understanding every failure this paper documents.

A. SUPACE

The Supreme Court introduced the Supreme Court Portal for Assistance in Court Efficiency (SUPACE), an AI-driven platform developed to support judges in managing complex caseloads. SUPACE is designed to analyse vast quantities of case records and identify legally relevant material, extract key precedents with remarkable speed, generate concise case summaries and spotlight pertinent issues, and organise documents to reduce the time judges spend on routine research and document review. The former Chief Justice confirmed that SUPACE will not spill over into judicial decision-making and is restricted to data collection and analysis [7].

B. SUVAS

A persistent structural challenge within the Indian judiciary arises from the country's extraordinary linguistic diversity. Although the Constitution recognises 22 scheduled languages, judicial discourse at the appellate level continues to be dominated by English, while trial-court proceedings take place in regional languages. This linguistic asymmetry generates a systemic disconnect, as the judgments that shape rights and obligations remain inaccessible to the citizens they affect.

SUVAS (Supreme Court Vidhik Anuvaad Software) was conceived precisely to address this linguistic deficit. Developed as an AI and machine-learning driven translation platform trained on domain-specific legal corpora, SUVAS institutionalises multilingual accessibility to Supreme Court judgments. In 2023 alone, SUVAS enabled the translation of approximately 36,000 Supreme Court judgments into 19 Indian languages [7]. Another transcription AI model used is TERES, which uses the BHASHINI platform for multilingual support [7].

C. LegRAA

The Legal Research Analysis Assistant (LegRAA) is a Generative AI-based system designed under the e-Courts framework to transform how judicial actors conduct research and prepare case materials. LegRAA rapidly analyses large volumes of legal documents including pleadings, judgements, and statutory materials, generates structured outputs such as briefs, issue-based summaries, and precedent lists, and draws on a corpus of over 36,000 Supreme Court judgments to identify key facts and surface relevant authorities within seconds [7].

D. e-Filing AI

The Supreme Court of India has initiated a significant pilot programme in collaboration with IIT Madras to test the use of AI and machine learning for automated defect detection in filings. The pilot project trains AI models on thousands of past filings to recognise recurring patterns of defects, extracts relevant metadata automatically at the time of e-filing, and aims to significantly reduce scrutiny time and minimise human error. Although still in the testing phase, this represents a major institutional step toward AI-driven quality control in India's judicial workflow [7].

E. Adalat AI

Recently, the Kerala High Court has also mandated the use of Adalat AI for all the courts in its jurisdiction for recording witness depositions. Adalat AI is a transcription tool used for converting speech to text in real time, thereby reducing delays caused by manual typing [7].

F. CMAPS

The Crime Mapping, Analytics and Predictive System (CMAPS) was a web-based decision support platform jointly initiated by Delhi Police and ISRO-ADRIN under a December 2015 MoU. Designed to spatially map and analyze crime data across parameters such as region, frequency, and type, CMAPS aimed to enhance insights into criminal behaviour for strategic policing. Key facts:

- ADRIN was responsible for developing analytical tools and the application; Delhi Police was tasked with infrastructure, data provisioning, and capacity building
- Planned for completion by December 2018 in four phases, the system remained incomplete
- Integration with CCTNS was not achieved, limiting data scope, and advanced modules were not implemented
- The CAG Report No. 15 of 2020 found significant implementation failures and concluded that "lack of appreciation of technology was evident throughout the project" [8]

G. Trinetra, Trinetra 2.0 and CrimeGPT

Trinetra is an advanced AI-based surveillance and criminal intelligence platform developed by Staqu Technologies in collaboration with the Uttar Pradesh Police. The system integrates facial recognition, voice recognition, and audio analytics with widespread CCTV networks, and is supported by a digitised database of over 900,000 criminal records, enabling rapid identification, history retrieval, and pattern analysis of suspects. Trinetra 2.0, also called CrimeGPT, was launched in March 2024 and extends this capability to generative AI-based querying; officers can ask questions in natural language about any of the 900,000 individuals in the database and receive responses including gang associations and criminal histories [9]. (See Appendix A for additional state-level AI tools including ABHED, PAIS, and TSCOP.)

It is worth noting that the failures documented in this paper did not come from SUPACE, LegRAA, or any of the purpose-built systems listed above. They came from general-purpose tools, primarily ChatGPT, introduced informally by individual judges, lawyers, and government officers without training, disclosure, or verification. However, this does not mean the purpose-built systems are safe. CMAPS was built specifically for Delhi Police and still embedded five documented forms of bias. Trinetra was built specifically for UP Police and still runs on a database of 9,00,000 records with no published bias audit. The purpose-built systems are larger, more deeply integrated, and far harder to challenge than a judge using a free chatbot. But the institutional adoption of these specialised tools has

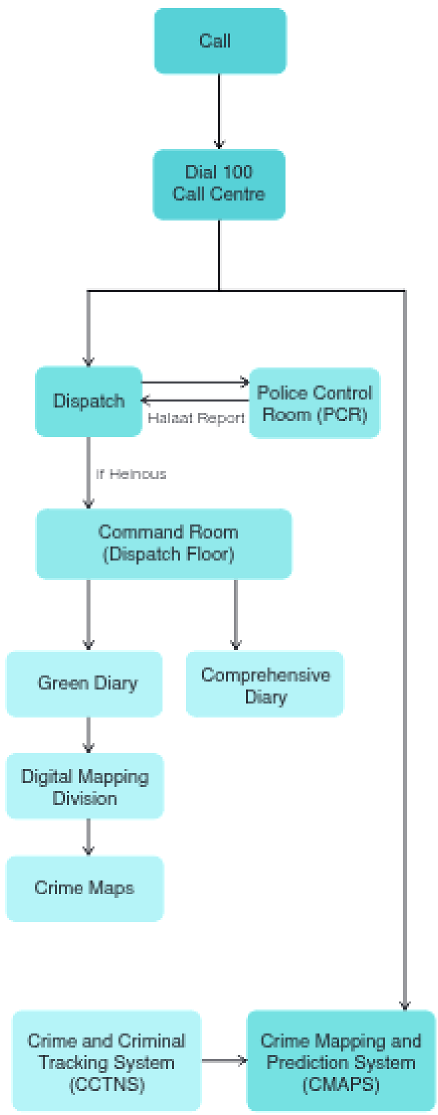

Fig. 1. CMAPS data flow from Dial 100 call centre to crime mapping output. Source: Marda and Narayan, ACM FAccT 2020 [12].

created a cultural permission structure through which general-purpose AI tools are now entering the system.

III. What Existing Research Has Found

The deployment of these systems has not gone unstudied. Three bodies of research are directly relevant to this paper.

A. *Marda and Narayan on CMAPS (FAccT 2020)*

Through direct observation of Delhi Police's call centre operations, Marda and Narayan identified five structural problems with how CMAPS generates and uses data [12].

1. Historical Bias: Law enforcement has a long-standing history of selective surveillance that disproportionately targets marginalised groups based on caste, class, gender, and religion. When this historical prejudice is fed into predictive models, the technology does not eliminate bias. It formalises and actively embeds these subjective human flaws directly into the system's data.

2. Representation Bias: The primary datasets powering CMAPS, emergency calls and historical police records, inherently under-represent affluent demographics. Because the vast majority of recorded emergency calls originate from slums and lower-income areas, the system automatically predicts higher crime probabilities in these specific zones. This creates a feedback loop: heightened algorithmic scrutiny leads to increased police deployment in marginalised neighbourhoods, which in turn generates more arrest data used to justify further surveillance.

3. Measurement Bias: The spatial mapping accuracy of the system heavily favours organised, privileged neighbourhoods over temporary or informal shanty settlements. Vulnerable demographics, such as women who largely remain indoors, often cannot provide precise addresses when reporting a crime. Officers map these complex locations inaccurately at the nearest police station, heavily skewing the spatial distribution of the city's actual crime hotspots.

4. Direct and Indirect Algorithmic Discrimination: The CMAPS software contains explicit mapping layers that allow police to actively filter for immigrant colonies and minority settlements. This direct discrimination is rooted in the prejudicial assumption that the mere existence of these communities inherently correlates with higher crime rates. Poor data collection in shanty towns also leads to clustered, inaccurate crime hotspots, effectively institutionalising the over-policing of vulnerable populations.

5. Hard-Coding Human Arbitrariness: The categorisation of a reported crime relies heavily on arbitrary, on-the-fly interpretations of call takers and dispatch officers. These highly subjective human interpretations are forced into 130 rigid, pre-defined categories on a standardised form, stripping away crucial context. Once this flawed, informal human judgment is digitised, the algorithm treats it with absolute mathematical certainty, transforming arbitrary social biases into rigid technological rules.

B. *Girhepuje et al. on Bail Prediction Fairness (AI and Law Symposium, 2023)*

While Marda and Narayan showed that the data feeding into India's policing AI is biased, Girhepuje and colleagues showed that this bias does not stop at policing. It propagates directly into judicial AI systems. Their study examined whether AI models trained on Indian legal data produce fair bail predictions [13].

They sampled 10,000 cases from the Hindi Legal Documents Corpus (HLDC), a corpus of over 900,000 Hindi legal documents. The dataset's own composition revealed a structural problem before any model was even trained: 64% of cases were bail-denied and only 36% were bail-granted. This means any model trained on this data begins with a built-in inclination toward denial. The basic characteristics of the corpus are shown in Table I.

TABLE I
TOKEN COUNTS FOR ORIGINAL CASE TEXT (FACTS AND ARGUMENTS SECTION)

Metric	Full Text	Bail Granted	Bail Denied
Mean	344.7	395.9	315.5
Median	324	380.5	300
Min	4	4	5
Max	1233	1233	1193

Source: Girhepuje et al. (2023) [13].

In plain language: bail-granted cases have significantly longer documented fact sections (mean 395.9 tokens) than bail-denied cases (mean 315.5 tokens). A model trained on this pattern learns to associate sparse documentation with denial, and sparse documentation disproportionately characterises cases of defendants from marginalised communities who have less access to legal support.

The researchers then trained a decision tree classifier using topic modelling to extract features from case text, and evaluated it using a counterfactual fairness framework, testing whether changing only the religion of the defendant, while keeping all other facts constant, would change the model's prediction. The result was an overall fairness disparity of **0.237** between Hindus and Muslims. A Muslim defendant and a Hindu defendant with identical legally relevant facts would receive materially different bail predictions from this model, purely because of religious community markers embedded in the training data [13]. The fairness gap across crime types is shown in Table II.

TABLE II
FAIRNESS GAP ON DENIAL OF BAIL ACROSS CRIME TYPES

Crime	Fairness Gap
Murder	0.071
Theft	0.48
Dowry	0.38
Drugs	0.054
Overall	**0.237**

Source: Girhepuje et al. (2023) [13].

This number is not abstract. NCRB 2022 data shows that Muslims constitute 19.3% of India's undertrial population despite being 14.2% of the general population. SC, ST, and OBC communities together make up 66% of undertrials, a share that has remained above 60% for 25 consecutive years [15]. An AI system trained on this data does not just reflect historical bias. It amplifies it, and gives it the false authority of a mathematical score.

C. The Supreme Court's Own Acknowledgement – Chapter 4 of the White Paper (2025)

The risks documented by researchers are not unknown to India's highest judicial authority. The Supreme Court's own Centre for Research and Planning, in its *White Paper on Artificial Intelligence and the Judiciary* (November 2025), dedicated an entire chapter to AI risks in the judiciary. It identified six categories: overreliance on unverified outputs and diminished human judgement; fabrication of cases and hallucination, including fake citation incidents; evidence tampering and deepfakes; algorithmic bias and discrimination; intellectual property infringement; and breach of confidentiality and privacy [7].

On hallucination specifically, the White Paper cited Stanford RegLab's finding that even specialist legal AI tools hallucinate between 17% and 33% of the time. General-purpose tools like ChatGPT hallucinate on legal queries between 58% and 82% of the time [14]. On algorithmic bias, the White Paper stated that AI systems "invariably inherit biases embedded in representations of reality encoded in raw training data... these biases can manifest in the form of racial, gender, ethnic, or cultural leanings, and can extend to other biases concerning socioeconomic status, religion, and more" [7].

The significance of this is straightforward: the Supreme Court of India, in its own official publication, has documented the exact failure modes that the ten cases in this paper exemplify. The problem is not that the risks are unknown. The problem is that acknowledged risk has not translated into binding regulation, mandatory safeguards, or accountability mechanisms. That gap is what this paper addresses.

D. What This Paper Does That Others Do Not

Marda and Narayan showed policing AI in India embeds historical discrimination. Girhepuje et al. showed judicial AI trained on Indian data propagates religious bias, with a measured disparity of 0.237. The Supreme Court White Paper acknowledged hallucination and algorithmic bias as institutional risks. Each of these is a significant contribution. But none of them has examined all four stages of India's criminal justice and judicial pipeline, policing, arrest, bail, and verdict, as one connected system, using ten documented real cases from 2020 to 2026 to show how failures at each stage compound into the next. The ten cases in this paper are selected because each one provides empirical evidence of a specific failure at a specific stage of the pipeline. When read in sequence, they show that a person moving through India's justice system today encounters AI at every stage, and that at every stage, the same three things are missing: mandatory disclosure, a verification protocol, and accountability.

IV. THE FOUR-STAGE PIPELINE

The ten cases in this paper are not ten separate problems. They are evidence of failures at four connected stages of one pipeline through which a person passes when they encounter India's justice system.

Stage 1 – Policing and Surveillance: This is where the pipeline begins. AI systems at this stage determine which areas are flagged as high-crime, which individuals are placed under surveillance, and which communities are disproportionately monitored. As Marda and Narayan showed, the data feeding these systems is already contaminated by decades of biased policing.

Delhi Safe City Project (2025):

- Initiated under the Nirbhaya Fund, originally meant for women's safety, but expanded into a general city-wide surveillance infrastructure without any new legal authorisation
- 2,100 AI-enabled cameras now live, with 15,000 existing cameras already integrated into the C4I network
- The FRT database being used has grown from 20,000 to 3,00,000 suspect profiles with no publicly disclosed legal basis for this expansion
- One database category is labelled "rabble-rousers and miscreants," a term with no formal legal definition, no court order required for entry, and no exit process
- No Privacy Impact Assessment was conducted before deployment [16]
- Protest footage has been used to populate this criminal suspect database, meaning exercising the constitutional right to protest can result in permanent profiling

Stage 2 – Arrest and Identification: When a policing AI flags a person, the next stage is identification and arrest. In India, this is dominated by Facial Recognition Technology operated by Delhi Police. The threshold for a positive match is 80% similarity, a threshold with no scientific basis, no independent validation, and no legal grounding. When this stage produces a wrong identification, the person arrested gets a FIR filed against them, which goes into CCTNS, the same database from which every downstream AI system learns.

Ratan Lal Murder Case, Northeast Delhi (2020) [1]:

- FIR 60/2020 arose from communal violence during anti-CAA protests on February 24, 2020
- 29 people were accused; 27 have since been granted bail and one discharged, indicating the identifications largely could not survive legal scrutiny
- Ali, a Muslim resident of Chand Bagh, was arrested purely on the basis of a facial recognition match and spent over four and a half years in pre-trial detention without a completed trial
- Mohammed, another accused in the same FIR, spent two years detained after four rejected bail applications while his wife was pregnant and his business collapsed
- The AI tool used (AI Vision by Innefu Labs) matched side-profile and rear-profile images from poorly lit alleys, conditions the vendor himself admitted degrade accuracy
- Defence counsel found that the person "identified" as Mohammed was at least five inches taller, had different hair, different footwear, and a different number of shirt pockets
- No Test Identification Parade was conducted for any of the 50-plus accused, despite being mandatory under the BPR&D national SOP
- Over 80% of the riot cases investigated through FRT have since resulted in acquittals or discharges

Stage 3 – Bail and Pre-Trial: Once arrested, a person's first legal question is whether they will be detained or released pending trial. In India, 75.8% of all prisoners are undertrials, people who have not been convicted of anything [15]. AI has entered bail jurisprudence in India through two documented pathways: prediction models carrying the 0.237 religious disparity documented by Girhepuje et al., and through the direct courtroom use of ChatGPT in bail orders. The output of a Stage 2 arrest, the FIR, the charge, the detention record, is what a Stage 3 bail decision must evaluate. If that arrest was wrong, the bail evaluation begins from a corrupted premise.

Jaswinder Singh Bail Order, Punjab and Haryana High Court (March 2023) [17]:

- Justice Anoop Chitkara of the Punjab and Haryana High Court denied bail to Jaswinder Singh, accused of assault causing death
- The judge queried ChatGPT: "What is the jurisprudence on bail when the assailants are assaulted with cruelty?"
- The AI's response, that courts are "less inclined to grant bail" in cruelty cases, was reproduced verbatim in the official bail order
- ChatGPT does not retrieve legal information. It predicts the next most likely word. It has no knowledge of Indian bail jurisprudence, no awareness of the specific facts of the case, and no accountability for what it outputs
- No disclosure was made to the accused that AI had influenced the bail order
- No training at the time required judges to flag or verify AI use in live proceedings [7]

Stage 4 – Judicial Decision and Verdict: The final stage is the judicial proceeding itself. This is where seven of this paper's ten cases are concentrated, because this is where AI failure has proliferated most rapidly and most visibly in India between 2023 and 2026. At this stage, generative AI tools have been used to fabricate legal citations in tribunal orders, High Court assessments, trial court judgments, and Supreme Court petitions. Three of the most consequential cases are presented here; additional documented cases are provided in Appendix A.

Gummadi Usha Rani v. Sure Mallikarjuna Rao, Andhra Pradesh Trial Court (August 2025) [4]:

- A trial court dismissed defendants' objections in a property dispute by citing four Supreme Court judgments
- All four were found to be AI-generated and non-existent by the Andhra Pradesh High Court
- The High Court still affirmed the decision on merits, stating that "mere mentioning of incorrect or non-existent rulings cannot be a ground to set aside the order"
- The Supreme Court intervened on February 27, 2026, stayed the order, and declared that relying on AI-generated fake judgments is not an error in decision-making; it is misconduct

KMG Wires v. National Faceless Assessment Centre, Bombay High Court (October 2025) [3]:

- The National Faceless Assessment Centre issued a Rs. 27.91 crore tax assessment citing three judicial decisions

- All three were found to be non-existent when the taxpayer's counsel attempted to locate them
- The Bombay High Court quashed the entire assessment
- The court observed: "In this era of Artificial Intelligence, one tends to place much reliance on the results thrown open by the system. However, when one is exercising quasi-judicial functions, such results are not to be blindly relied upon"

Nationwide AI Citation Crisis, Supreme Court of India (February 2026) [5]:

- Chief Justice Surya Kant publicly described AI-generated fake citations as an "alarming" and system-wide crisis
- A petition before the court cited a wholly non-existent case titled "Mercy v. Mankind," a case that has never existed in any jurisdiction anywhere in the world
- In Justice Dipankar Datta's courtroom, a series of AI-fabricated judgments were cited across multiple different matters
- The Supreme Court issued notices to the Attorney General, the Solicitor General, and the Bar Council of India
- A new and more dangerous escalation was also identified: AI is now fabricating passages within real cases, meaning checking whether a case exists is no longer sufficient; the quoted text must also be verified against the original source

V. How the System Connects

These four categories are not four separate problems. They are four stages of one journey that a person takes through India's justice system, and at each stage, an AI system is waiting.

A person first encounters AI at the policing and surveillance stage, where they may be flagged by a system that cannot define its own criteria. The Safe City project uses a database that includes "rabble-rousers and miscreants," a category with no legal definition. The CMAPS system maps crime to areas based on who calls the emergency line, not where crime actually occurs. A person from a marginalised community, in an over-policed neighbourhood, is more likely to be flagged, not because of what they have done, but because of where they live and who they are.

If flagged, they encounter AI at the arrest and identification stage, where a facial recognition tool with an arbitrary 80% threshold decides if they are a suspect. Ali was identified from a side-profile image taken in a poorly lit alley. Mohammed was matched to a man five inches taller than him, wearing the same common outfit. Both were arrested. Both spent years in pre-trial detention. Neither was convicted.

If arrested, they encounter AI at the bail stage, where a judge may consult a chatbot without any disclosure to the accused. The Jaswinder Singh case shows that even the most consequential pre-trial decision, whether a person walks free or is held in detention, can now be influenced by a language model that predicts words rather than retrieves law. The accused had no way to know ChatGPT was consulted. They had no way to challenge it.

If held for trial, they encounter AI at the judicial decision stage, where the judgment that determines their fate may contain citations that do not exist. From the Karnataka trial court to the Andhra Pradesh property dispute to the Supreme Court itself, the pattern is the same: AI-generated legal authority, submitted without verification, deciding real outcomes for real people.

At each stage, the output of the previous AI system has shaped what happens at the next one. The FRT arrest produced the FIR that fed the bail decision. The bail denial produced the detention record that informed the trial. The trial used AI research that was never verified. Each failure compounded the one before it.

The person in this system has no way to know that AI was involved at any stage. They cannot challenge a tool they do not know was used. They cannot verify citations they were never shown. They cannot contest a database entry they do not know exists. The invisibility of AI in this pipeline is not a side effect. It is its most dangerous feature. Every safeguard, the right to challenge evidence, the right to know the basis of a decision, the right to a fair trial, depends on the person knowing what was used against them. When AI is invisible, those rights become hollow.

VI. Two Technical Failure Modes the Governance Literature Has Not Named

The cases in the preceding sections are easy to frame as governance failures: no disclosure, no verification, no audit. That framing is correct but incomplete. Two of the failure modes documented here are not just policy problems. They are technical design problems that governance alone cannot fix.

A. The Growing Database Problem: Why the Same Threshold Becomes Less Reliable Over Time

Delhi Police treats any facial recognition match above 80% similarity as a positive identification. This number sounds reassuring. But a confidence score from a facial recognition system is not a probability of guilt. It is a mathematical distance between two face-embedding vectors. That is a very different claim from "this person was at the scene."

The deeper problem is how this threshold behaves as the database grows. When you search one face against a database of 20,000 profiles at a 1% false positive rate, you get roughly 200 incorrect flags. When Delhi Police expanded their FRT database to 3,00,000 suspect profiles by Independence Day 2025 [16], that same false positive rate produces roughly 3,000 incorrect flags per search. The threshold did not change. The algorithm did not change. But the number of people wrongly identified multiplied by fifteen.

This is called the base rate problem, and it is one of the most documented failure modes in applied machine learning. Delhi Police had an obligation to either adjust the threshold when the database expanded, or to disclose to courts that the reliability of each identification had materially decreased. They did neither. No court was told. No accused was informed. Ali's case, and the 80% acquittal rate across the 750 riot cases, is consistent with exactly what base rate calculations would predict from a 1:N search run against a database of this size, at this threshold, without recalibration [1].

B. The Absent Feedback Signal: Why These Systems Cannot Self-Correct

Every machine learning system improves through the same basic mechanism. It makes a prediction, receives a signal about whether that prediction was correct, and adjusts. Without this feedback, a model cannot improve. It can only repeat the patterns it learned from its original training data, including the same errors.

India's criminal justice AI systems have no feedback loop.

Ali was arrested based on an FRT match. He spent four and a half years in detention. He was never convicted. That outcome is a clear signal that the original identification was wrong. But this signal never reached the model. The acquittal is recorded in the court's case management system, not in the FRT system's training database. These two systems do not communicate. There is no protocol requiring Delhi Police to report wrongful FRT identifications back to the developer. The model is formally described as operational. It is functionally frozen at the state of its original training data.

The same absence appears in the judicial AI cases. When the Karnataka High Court escalated the trial court judge's use of fabricated citations, that correction was a legal event [19]. It was not a training signal to ChatGPT. The next judge who queries the same tool on the same legal question will receive the same hallucinated output. The Supreme Court can declare AI hallucination misconduct. That declaration cannot reach the model. India's courts can catch the errors after they happen. They cannot prevent the same errors from happening again.

This is not just a governance failure. It is a technical design problem. A system with no feedback signal is not merely unaccountable. It is architecturally incapable of improvement.

VII. Why India Is Particularly Vulnerable

India passed the Digital Personal Data Protection Act in 2023. It does not mention algorithmic systems. It does not require fairness audits. It does not classify any AI application as high risk. The EU has the AI Act, which explicitly places predictive policing, biometric identification, and judicial AI in its highest-risk category. The US has executive orders on AI in criminal justice. India has guidelines and white papers, none of which are binding. The Supreme Court's own November 2025 White Paper gives excellent recommendations. None of them are law. A recommendation that carries no penalty for non-compliance is indistinguishable from no recommendation at all. The legal vacuum is not a gap, it is the floor on which every failure in this paper was built.

CCTNS, India's national crime database, is built from FIR data. An FIR records who police decided to arrest, not who actually committed a crime. In India, Dalits, Muslims, and Adivasis are disproportionately arrested, not because they commit more crime, but because they are disproportionately policed. NCRB 2022 confirmed this numerically [15]. Any AI trained on CCTNS data learns this arrest pattern as if it were a pattern of criminality. The model does not know the difference. It treats historical policing decisions as ground truth. This is not a fixable training data problem, it is a structural contamination that would require dismantling decades of discriminatory enforcement records before a fair model could be trained on them.

Section 8(1)(h) of the RTI Act allows law enforcement to withhold information that might impede the investigation or prosecution of offenders. Delhi Police has used this exemption to block every meaningful RTI query about CMAPS and FRT. Marda and Narayan filed 13 RTIs and received no usable information [12]. The Internet Freedom Foundation's RTIs about the Safe City project produced similarly opaque responses [16]. A system that cannot be seen cannot be challenged. This is the direct reason why none of the ten cases in this paper were caught before harm was done, the tools operated entirely outside public view, and the RTI framework actively protected that invisibility.

Staqu Technologies built Trinetra for UP Police, PAIS for Punjab Police, ABHED for Rajasthan Police, and components of other state systems. The same algorithm, with the same potential flaws, is running simultaneously across multiple states under different names. Nobody is coordinating an audit across all of them. No state is required to disclose which vendor built their criminal justice AI. If Staqu's underlying model carries a bias, and there is no published audit suggesting it does not, that bias is not isolated to one state. It is a quietly replicated nationwide standard, protected by vendor confidentiality and RTI exemptions simultaneously.

The national Sub-Inspector training syllabus, as of 2024, allocated one period to AI and one period to facial recognition out of 2,620 total instructional hours. The Dwarka Cyber Training Division ran zero documented FRT training sessions in all of 2024. The judges who consulted ChatGPT in bail orders, in Karnataka [19], and in Andhra Pradesh [4] received no formal training on how language models produce output. They were not told what hallucination means. They were not told that an 80% confidence score is a vector distance, not a verdict. They used powerful tools they did not understand because nobody required them to understand before using. That is not individual negligence. That is institutional failure.

VIII. Proposed Interventions

The failures in this paper were not random. Each one happened inside a specific gap: no disclosure requirement, no verification protocol, no feedback mechanism, no audit obligation, no training mandate. The interventions below address those gaps directly, not generically. Each one has a governance layer that tells institutions what they must do, and a technical layer that tells AI developers what they must build. Both layers are necessary, because a rule with no technical enforcement is just a recommendation, and India has enough of those already.

Intervention 1: Citation Verification as a Pre-Filing Obligation (Governance + Technical).

On the governance side: every AI-assisted citation submitted to any Indian court must be cross-verified against an authorised primary database, Manupatra, SCC Online, or the official e-SCR, before filing. The Bar Council of India issues a binding circular under its existing Professional Standards Rules. Violation is treated the same as submitting fabricated evidence. This is not new law, it is an extension of an existing professional obligation.

On the technical side: any AI tool sold or marketed for legal use in India must integrate a real-time citation verification layer before outputting any case reference. The AI's output must be cross-checked against an authorised Indian legal database before it is shown to the user. If the cited case does not exist in that database, the tool must flag it as unverified. The Stanford study by Magesh et al. (2025) already shows that Retrieval-Augmented Generation tools hallucinate at 17–33% instead of 58–82%, because they retrieve from real databases before generating [14]. Making this mandatory for Indian legal databases is technically straightforward. This one change would have prevented the Bengaluru ITAT Buckeye Trust case [2], the Karnataka trial court incident [19], the Andhra Pradesh property dispute [4], both Bombay High Court cases [3], [20], and the Supreme Court's nationwide citation crisis [5].

The limitation is that a lawyer under time pressure can sign a verification declaration without actually verifying. The technical layer directly addresses this by making verification automatic rather than voluntary.

Intervention 2: Mandatory AI Disclosure in Every Court Record (Governance).

This directly addresses the invisibility problem documented in Sections **??** and **??**.

Every court record, order, judgment, submission, or assessment that involved AI assistance at any stage must state the name of the tool, the version used, the task it performed, and the verification steps taken. This applies to judges, advocates, and government officers equally. The Supreme Court issues a Practice Direction applying this to all courts below it. The Bar Council of India issues a parallel circular. The Ministry of Home Affairs issues a separate circular for law enforcement agencies.

Mandatory disclosure creates an audit trail that does not currently exist. Without it, the scale of AI-driven harm across India's courts is permanently unknowable. With it, appellate courts can identify patterns, journalists can investigate, and researchers can compile the error rates that no one is currently tracking.

The limitation is that disclosure of AI use does not guarantee the output was verified. This is why Intervention 1 must accompany it.

Intervention 3: Counterfactual Fairness Requirement for Bail and Risk Models (Technical).

This directly addresses the 0.237 fairness disparity documented by Girhepuje et al. [13].

Any AI model used to inform bail decisions in India must satisfy counterfactual fairness, a formal mathematical property established by Kusner et al. (NeurIPS 2017) [21]. Counterfactual fairness requires that if a defendant's religion or caste were changed while holding all legally relevant features constant, the model's output must not change. This is stricter than standard group fairness metrics because it tests the individual outcome, not the average. In India's context, where historical data is contaminated by community-specific over-policing, counterfactual fairness directly targets the contamination at the model's decision boundary.

Any developer building an AI system for use in Indian bail proceedings, including under the e-Courts Phase III framework, must certify counterfactual fairness compliance before deployment. MeitY's existing AI Governance Guidelines (2025) already recommend fairness testing. A statutory instrument under the IT Act 2000 makes this certification mandatory rather than advisory.

The limitation is that counterfactual fairness certification requires access to protected attribute data during testing, which raises its own privacy considerations that must be addressed in the implementing regulation.

Intervention 4: Demographic Disparity Audit Before Any CCTNS-Trained System Goes Live (Governance + Technical).

This directly addresses the wrongful arrests in the Ratan Lal Murder Case [1] and Delhi's Safe City Project [16], and the structural contamination described in Section VII.

On the governance side: any AI system trained on CCTNS data must produce a demographic disparity report, comparing predicted risk distributions against population base rates by religion, caste, and district, before deployment. If the flagging rate for any protected demographic group exceeds their share of the general population by more than a defined margin, deployment is paused pending independent review. NCRB is the implementing body. Additionally, any expansion of a deployed FRT database must trigger a mandatory threshold recalibration, addressing the base rate problem described in the preceding section.

On the technical side: the disparity report must use Equalised Odds as the standardised fairness metric, meaning the system must be equally accurate across demographic groups in both positive and negative predictions. This is the same methodology Girhepuje et al. applied to Indian bail data, which means the regulatory standard is not theoretical but directly derived from peer-reviewed research on Indian legal datasets.

The limitation is that NCRB's own data collection reflects the same historical policing bias that contaminates CCTNS. The audit catches whether the model reproduces that bias, but cannot remove the bias from the training data itself.

Intervention 5: Mandatory AI Literacy Gate Before Any Judicial or Policing AI Tool Can Be Used (Governance + Technical).

This addresses the training gap described in Section VII and the automation bias documented across the Ratan Lal Murder Case [1], the Jaswinder Singh bail order [17], and the Karnataka trial court incident [19].

On the governance side: no judge may use an AI tool in live proceedings without completing a certified training module covering how LLMs generate text, what hallucination is and why it is structurally inevitable, and what verification steps are required before any AI output can be included in a court document. No police officer may operate an FRT system without completing a certified module on false positive rates, the 1:N identification problem, and the legal requirements for a valid identification. The National Judicial Academy designs and mandates the judicial module. Police Training Colleges integrate FRT accuracy training into all relevant syllabi. The Kerala High Court's July 2025 policy requiring AI training for all district judiciary staff is the model; it must become national.

On the technical side: AI tools deployed in judicial or policing contexts must include an onboarding module that cannot be bypassed. Before a user can access the tool's output, they must have completed and passed a basic verification skills assessment embedded in the tool itself. This is not surveillance of users. It is a technical gate ensuring minimum competence before use. A judge who has never been told what hallucination is should not be able to open a legal AI tool the same way a pilot who has never trained on an instrument cannot legally fly in cloud.

The limitation is that training mitigates automation bias but does not eliminate it. Structural pressure from caseload will continue to incentivise over-reliance on AI output. Intervention 1's mandatory verification layer is the structural counterweight.

Intervention 6: RTI Reform for Criminal Justice AI (Governance + Legal).

This addresses the opacity that allowed every system documented in this paper to operate without external scrutiny.

On the governance side: criminal justice AI systems must be carved out from the blanket law enforcement exemption under Section 8(1)(h) of the RTI Act. The following information must be made publicly accessible for any deployed system: the training data composition, the accuracy rates broken down by demographic group, the number of people flagged and subsequently acquitted, and the vendor contract terms. Parliament must amend Section 8(1)(h) to exclude algorithmic systems explicitly. An independent technical auditor, appointed under the DPDP framework, would have access to information that cannot be made fully public due to legitimate security concerns, ensuring scrutiny without compromising operational details.

There is no technical layer here because the RTI problem is purely legal. A technical audit framework cannot function if the law prevents access to the information an audit would require.

The limitation is that legislative reform takes time, and the systems documented in this paper are already operational. Interim measures, including the disclosure requirements under Intervention 2, provide partial mitigation while the statutory change is pursued. But the RTI exemption itself is the foundational obstacle. Marda and Narayan filed 13 RTI requests about CMAPS and received no useful information [12]. The IFF's RTIs about the Safe City project produced similarly opaque responses [16]. The Supreme Court's own White Paper acknowledged that opacity in law enforcement AI is a structural problem [7]. Guidelines and circulars cannot override a statutory exemption. Only Parliament can fix this.

IX. Discussion and Limitations

This paper has two honest limitations that any reviewer should be aware of.

First, most policing AI in India operates behind RTI exemptions. The analysis of bias in CMAPS, FRS, and Trinetra is based on documented external evidence: Marda and Narayan's fieldwork [12], IFF's RTI responses [16], and the Pulitzer Center investigation [1]. None of these sources had access to the actual training data, accuracy rates, or deployment logs of these systems. A proper technical audit would require legal access that currently does not exist. The RTI reform proposed in Intervention 6 is partly motivated by the fact that this paper could not access information that it needed to make stronger empirical claims.

Second, India does not yet have a deployed recidivism scoring system equivalent to COMPAS in the United States. The correctional AI node exists: Trinetra's Jarvis module in UP prisons is documented. But its direct algorithmic connection to bail or parole decisions is not yet fully verified through court records. The four-stage pipeline described in this paper is partly prospective at the correctional end. The point of documenting it now is to identify and name the structure before it is fully locked in, while intervention is still possible.

X. Conclusion

India's AI systems in criminal justice are failing at every stage of the pipeline. The failures are not random. They follow a pattern that concentrates harm on people who are already the most vulnerable and the least able to challenge a system that cannot be seen and cannot self-correct.

Six things need to change: mandatory citation verification with a technical enforcement layer, a disclosure requirement for every court record that touched AI, a counterfactual fairness test for bail prediction models, a demographic disparity audit before any CCTNS-trained system goes live, a compulsory training gate before any judge or officer can operate an AI tool, and a legislative amendment to the RTI Act that makes criminal justice AI auditable from outside. None of these require new technology. None of them require new courts. They require political will and institutional accountability, which are harder to build than any algorithm but far more durable once built.

India is building AI into its justice system faster than it is building the rules to govern it. The cases in this paper are not cautionary tales from the future. They are documented harms from the recent past. The question is not whether to use AI in the courts. That decision has already been made, informally, by individual judges with access to a free chatbot and a backlog that never ends. The question is whether India will choose to govern what it has already deployed, before the costs become irreversible.

Appendix

A. Additional AI Tools Deployed in Indian States

Alongside the systems described in Section II, several additional state-level tools are operational:

- **AI Saransh** – Developed by the National Informatics Centre; assists judges by generating concise summaries of legal pleadings, enabling faster comprehension of complex cases [7]
- **NyayKaushal** – Positioned as India's first e-Resource Centre and Virtual Court; integrates AI for virtual hearings, remote filings, and streamlined case management [7]
- **LESA** – An AI-powered chatbot introduced by the National Legal Services Authority that helps litigants track applications and navigate legal processes [7]
- **ABHED** – Staqu Technologies collaborated with the Rajasthan Police to introduce ABHED, an AI-enabled mobile application for criminal identity registration and information about missing persons. This technology has digitised more than 90,000 criminal records and has been instrumental in solving over 100 criminal cases [10]
- **PAIS** – The Punjab Police employed Staqu's technology to create the Punjab Artificial Intelligence System (PAIS), which facilitates searches across the FIR database, integrates biometric data, and assists in resolving high-profile cases. The system facilitates phonetic search to accommodate spelling discrepancies in names [10]
- **TSCOP** – The Telangana Police integrated advanced artificial intelligence into the TSCOP app, which facilitates real-time searches across vast criminal databases, integrates live facial recognition technology, and assists frontline officers in making rapid field decisions. The system utilises predictive analytics to forecast potential crime hotspots and proactively deploy patrol resources [11]

B. Additional Judicial Decision and Verdict Cases

The following cases from Stage 4 of the pipeline provide additional empirical evidence beyond the three presented in Section **??**.

Christian Louboutin SAS v. The Shoe Boutique, Delhi High Court (August 2023) [18]:

- The plaintiff's counsel submitted a ChatGPT-generated response as evidentiary material to establish the brand's global reputation
- Justice Pratibha M. Singh rejected it, holding that AI responses "cannot be the basis of adjudication of legal or factual issues in a court of law"
- Because the plaintiff won on independent grounds, no penalty was imposed for the AI submission; the precedent prohibits the practice without deterring it
- This was the first explicit judicial holding in India that LLM output cannot constitute the basis for adjudicating legal or factual issues

Bengaluru ITAT Buckeye Trust Case (December 2024) [2]:

- A Rs. 669 crore tax tribunal order was issued citing three Supreme Court judgments and one Madras HC ruling
- All four were entirely non-existent in any legal database
- The order was recalled within a week after the tax department's own representative caught the fabrication
- The Supreme Court's November 2025 White Paper specifically cited this incident as an early documented warning of AI hallucination entering quasi-judicial proceedings [7]

Karnataka Trial Court (January 2025) [19]:

- A trial court judge used ChatGPT to research precedents while drafting a judgment
- The AI-generated citations were either entirely invented or attributed legal propositions to cases that say something entirely different
- Neither party had cited those cases; the judge independently introduced AI-fabricated law into the proceedings without either party's knowledge
- The Karnataka High Court escalated the matter and proposed disciplinary action against the judge [7]
- This is the failure mode with no natural corrective mechanism: when a judge introduces fabricated authority independently, neither party can challenge it before the judgment is issued

Bombay High Court – Rs. 50,000 Costs for Unverified AI Citations (Early 2026) [20]:

- A litigant's counsel submitted written arguments containing non-existent case law generated by AI
- The Bombay High Court imposed costs of Rs. 50,000 and held that the duty to verify citations is absolute and cannot be delegated to an AI tool
- This is the first financial penalty in India specifically targeting unverified AI-generated citations submitted to a court

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